Tianle Wang

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Research interests: LLM Reasoning, Reinforcement Learning, Post-Training

Education

Purdue University West Lafayette, IN
Ph.D. in Computer Science *Aug. 2025 – Present*
Advisor: Prof. Abulhair Saparov, GPA: 4.0/4.0

University of California, San Diego La Jolla, CA
M.S. in Data Science *Sept. 2023 – Mar. 2025*
Advisor: Prof. Jingbo Shang, GPA: 4.0/4.0

Shanghai Jiao Tong University Shanghai, China
B.E. in Computer Science *Sept. 2019 – Jun. 2023*
ACM Honors Class, GPA: 3.7/4.3

Experience

Learning Machine, Inc. Palo Alto, CA
Member of Technical Staff Intern *May 2026 – Aug. 2026*

- Built an end-to-end, Rust-based web data pipeline over the full Common Crawl, processing raw WARC files into approximately 3T tokens of pretraining data.
- Applied document- and line-level heuristic filtering, exact and MinHash deduplication, and quality filtering, iterating on the pipeline with feedback from small proxy-model ablations.

Purdue University West Lafayette, IN
Research Assistant, advised by Prof. Abulhair Saparov *Aug. 2025 – Present*

- Developed **ScaleLogic**, a synthetic logical reasoning environment that independently controls proof depth and logical expressiveness (from implication-only rules towards first-order logic), to study RL scaling for long-horizon reasoning.
- Showed that RL compute follows a power law in proof depth, with the exponent rising from 1.05 to 2.60 as expressiveness grows, consistent across RL algorithms and model scales.
- Trained Qwen3-4B and 8B on the most expressive setting, improving average accuracy by over 10 percentage points across eight mathematical and general reasoning benchmarks.

University of North Carolina at Chapel Hill Chapel Hill, NC
Research Intern, advised by Prof. Huaxiu Yao *Jun. 2024 – Dec. 2024*

- Introduced **Anyprefer**, an agentic framework that synthesizes high-quality preference data for foundation models, and released a dataset of 58K preference pairs.
- Implemented a cooperative Markov game-based feedback mechanism between the target model and a tool-augmented judge model, jointly improving data quality and reward estimation accuracy.
- Improved alignment across 21 benchmarks, with average gains of 18.55% on natural language generation, 30.05% on medical image analysis, and 16.00% on visuo-motor control.

Massachusetts Institute of Technology, SLS Lab Cambridge, MA

Research Intern, advised by Prof. James Glass and Dr. Tianxing He *Jul. 2022 – Dec. 2022*

- Revealed the lack of robustness in PLM-based NLG metrics (e.g., BERTScore) with a synthetic-data evaluation system, identifying their biases and flaws and proposing mitigations.

University of California, San Diego, Shang Lab La Jolla, CA

Research Intern, advised by Prof. Jingbo Shang *Aug. 2021 – Dec. 2024*

- Proposed **WOT-Class** for weakly supervised open-world text classification, jointly improving class identification and document classification with over 20% macro-F1 gains.
- Built a benchmark comparing representation learning and prompting approaches for extremely weakly supervised text classification, analyzing key factors behind model performance.

Publications

*equal contribution

[1] **Tianle Wang**, Zhaoyang Wang, Guangchen Lan, Xinpeng Wei, Sipeng Zhang, GuanWen Qiu, Abulhair Saparov. "How Deep Can LLMs Learn to Reason? Expressiveness Is Key." *Preprint, 2026*.

[2] Yiyang Zhou*, Zhaoyang Wang*, **Tianle Wang***, et al. "Anyprefer: An Agentic Framework for Preference Data Synthesis." *ICLR 2025*.

[3] **Tianle Wang**, Zihan Wang, Weitang Liu, Jingbo Shang. "WOT-Class: Weakly Supervised Open-world Text Classification." *CIKM 2023*.

[4] Zihan Wang*, **Tianle Wang***, Dheeraj Mekala, Jingbo Shang. "A Benchmark on Extremely Weakly Supervised Text Classification: Reconcile Seed Matching and Prompting Approaches." *Findings of ACL 2023*.

[5] Guangchen Lan, Sipeng Zhang, **Tianle Wang**, et al. "MaPPO: Maximum a Posteriori Preference Optimization with Prior Knowledge." *NeurIPS 2026*.

[6] GuanWen Qiu, **Tianle Wang**, Abulhair Saparov. "Context Matters: Cross-Layer Decomposition of Sparse Autoencoder Features." *Under review, 2026*.

[7] Mengke Zhang, Tianxing He, **Tianle Wang**, et al. "LatticeGen: Hiding Generated Text in a Lattice for Privacy-Aware Large Language Model Generation on Cloud." *Findings of NAACL 2024*.

[8] Tianxing He, Jingyu Zhang, **Tianle Wang**, et al. "On the Blind Spots of Model-Based Evaluation Metrics for Text Generation." *ACL 2023*.

Skills

Languages: Python, C/C++, Rust, Bash

ML & Systems: PyTorch, Transformers, vLLM, SGLang, verl, FSDP, Ray, Slurm, RL post-training, multi-GPU training, pretraining data processing

Honors & Awards

Irving T. Ho Memorial Scholarship, one of 4 recipients at SJTU 2022

Zhiyuan Honorary Scholarship, top 5% at SJTU 2019 – 2022

Silver Medal, China National Olympiad in Informatics (NOI) 2017